



MACHINE LEARNING IN BIOINFORMATICS

Part 9: Neural Network Applications in Bioinformatics

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Neural Network Application in Bioinformatics



- Neural network have numerous applications in bioinformatics
- They are used in gene structure prediction, protein structure prediction, gene expression data analysis, ... Almost anywhere when you need to do classification.
- Here we specifically focus on applying neural networks to protein structure prediction (**secondary structure, solvent accessibility, disorder region, contact map**).

Outline

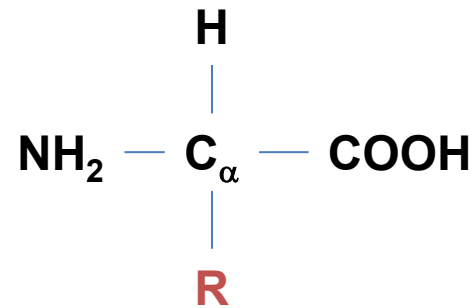


- 1. Proteins**
2. Secondary structure
3. Protein structure determination
4. Using neural networks for protein structure prediction
5. Predicting solvent accessibility, disordered region, contact map,

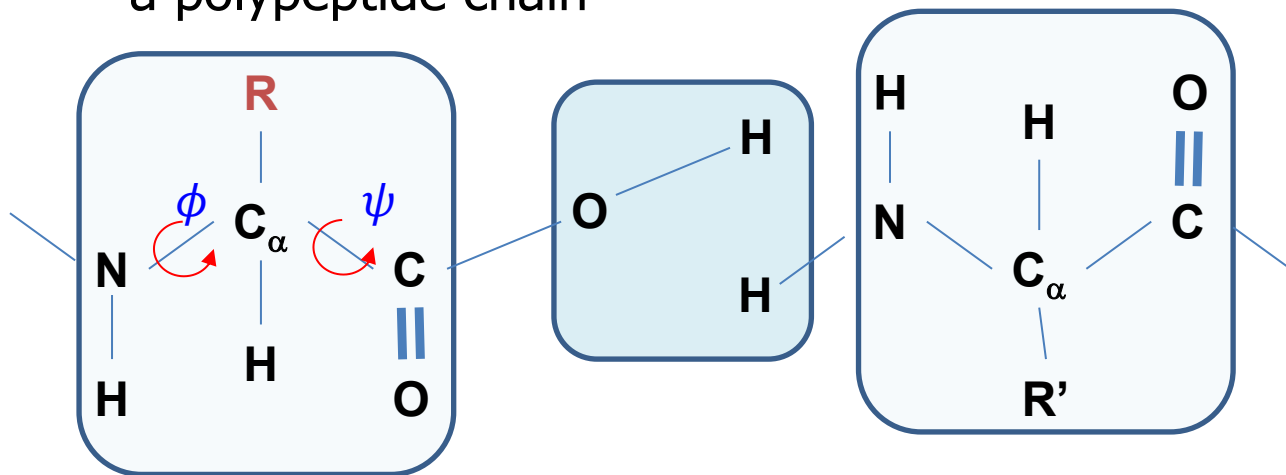
Proteins



- A protein is a chain of amino acids joined by peptide bonds
- The structure of an amino acid



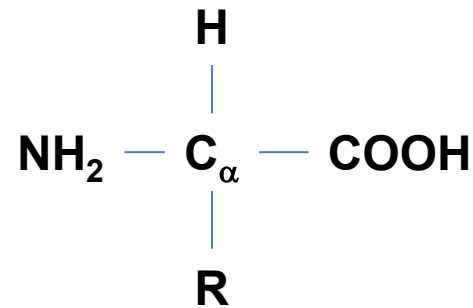
- amino acids are composed into a polypeptide chain



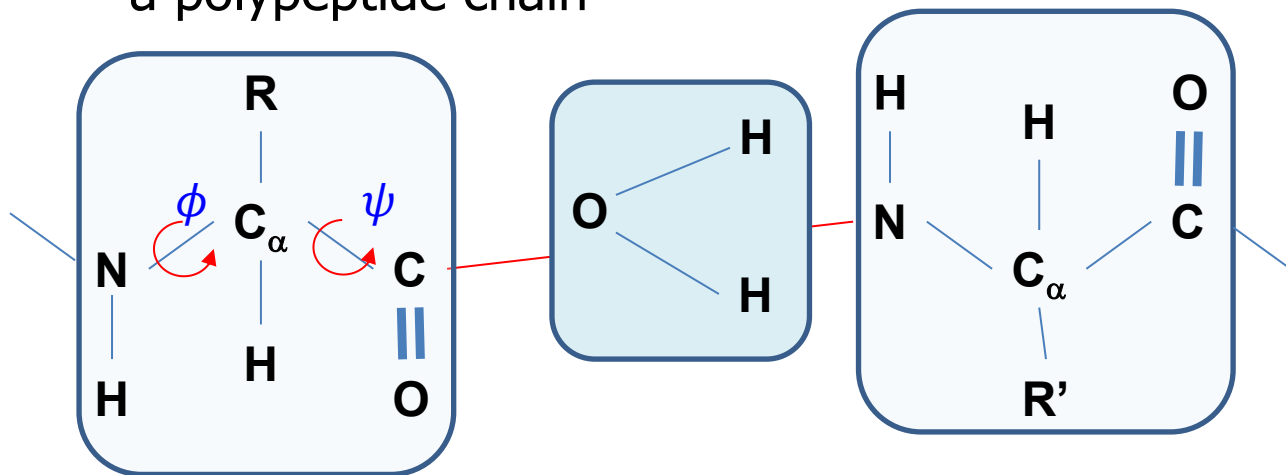
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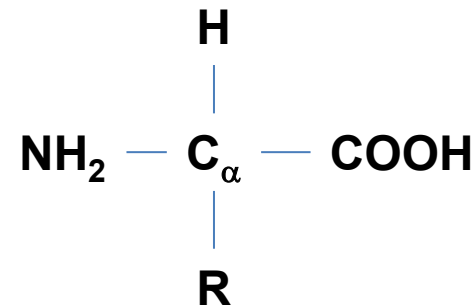
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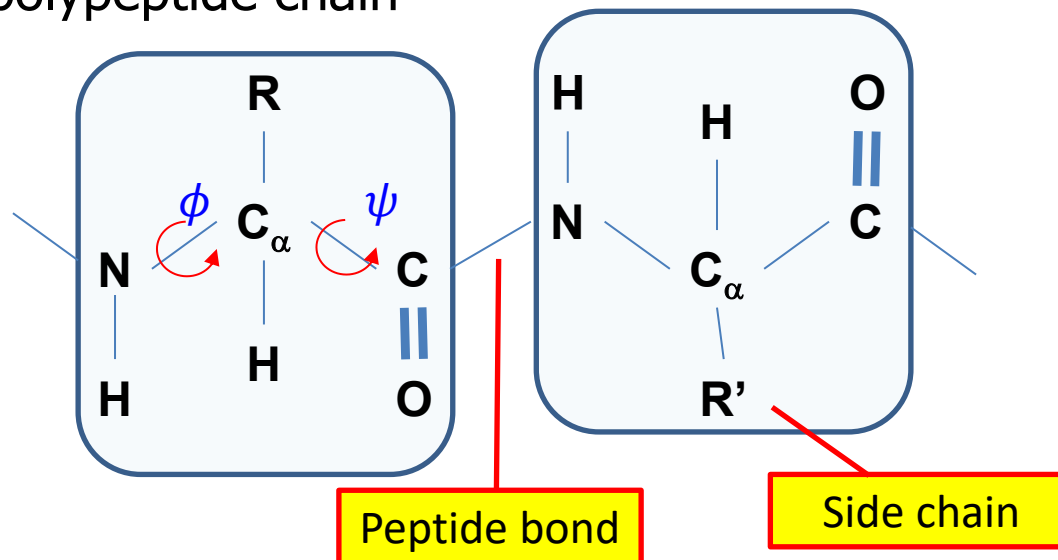
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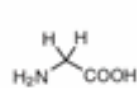


- N-C_α-C make up the backbone of the protein.
- Each amino acid has two rotational degrees of freedom ϕ and ψ
- The angle between C=O and N-H is always approximately 180°

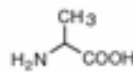
Amino Acids



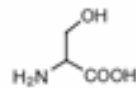
- <http://groundupstrength.wdfiles.com/local--files/amino-acids/amino-acids.gif>



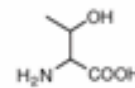
Glycine



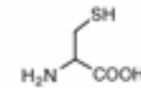
Alanine



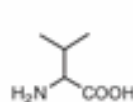
Serine



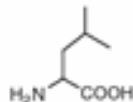
Threonine



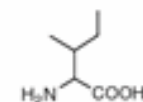
Cysteine



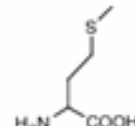
Valine



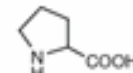
Leucine



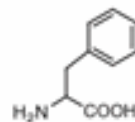
Isoleucine



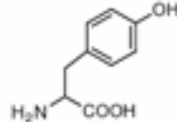
Methionine



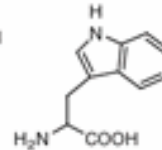
Proline



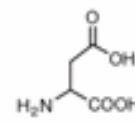
Phenylalanine



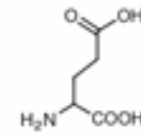
Tyrosine



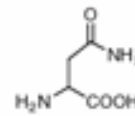
Tryptophan



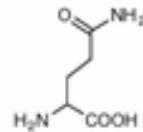
Aspartic Acid



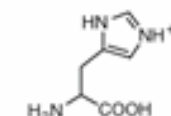
Glutamic Acid



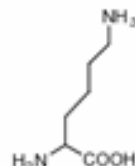
Asparagine



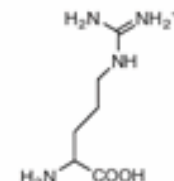
Glutamine



Histidine



Lysine



Arginine

Hierarchy of protein structure



- The **primary structure** is the chemical structure of the polypeptide chain(s) in a protein, i.e., its sequence of amino acid residues that are linked by peptide bonds.
- The **secondary structure** is folding of the molecule that arises by linking the C=O and NH groups of the backbone together by means of hydrogen bonds.
- The **tertiary structure** is the 3D structure of the molecule consisting of secondary structures linked by “looser segments” of the polypeptide chain stabilized (primarily) by sidechain interactions.
 - Protein shape determines most of its function
 - Experimental determination of protein structure via x-ray crystallography is hard and time consuming
 - We would like to determine the structure of a protein from its sequence
- The **quaternary structure** is the aggregation of separate polypeptide chains into the functional protein.

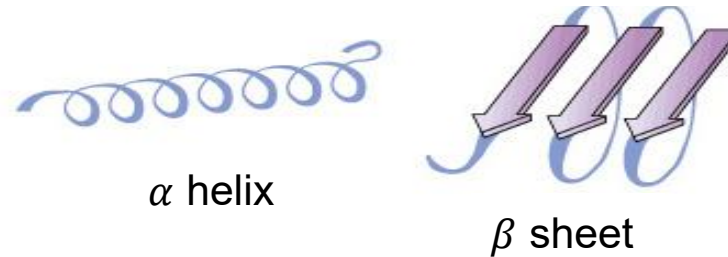
Four Levels of Protein Structure



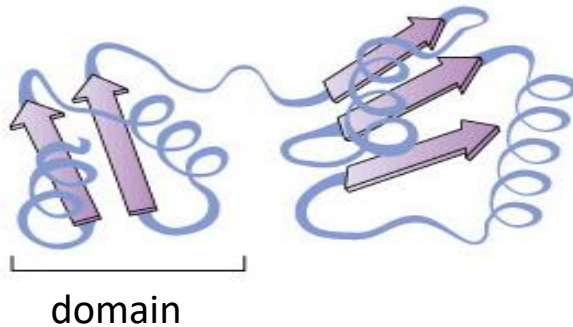
a) Primary structure

–Ala–Glu–Val–Thr–Asp–Pro–Gly–

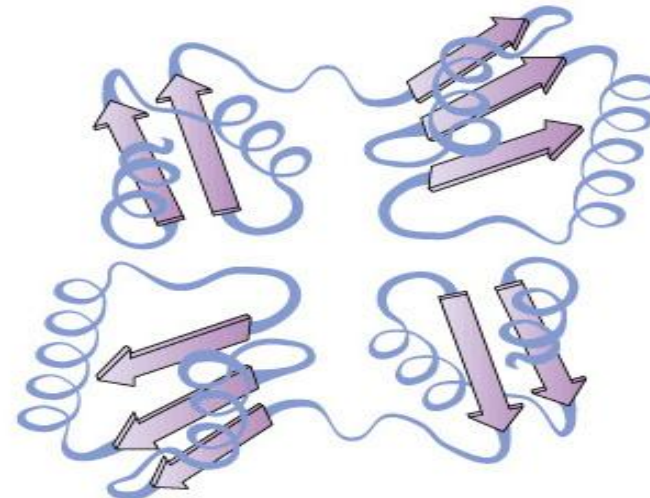
b) Secondary structure



c) Tertiary structure



d) Quaternary structure



Outline



1. Proteins
2. **Secondary structure**
3. Protein structure determination
4. Using neural networks for protein structure prediction
5. Predicting solvent accessibility, disordered region, contact map,

Secondary structure



- Determined by hydrogen bond patterns
- 3-Class categories:
 - α helix
 - β sheet,
 - loop (or coil)
- First deduced by Linus Pauling et al.
- α helix and β sheet correspond to specific choices of the ϕ and ψ angles along the chain

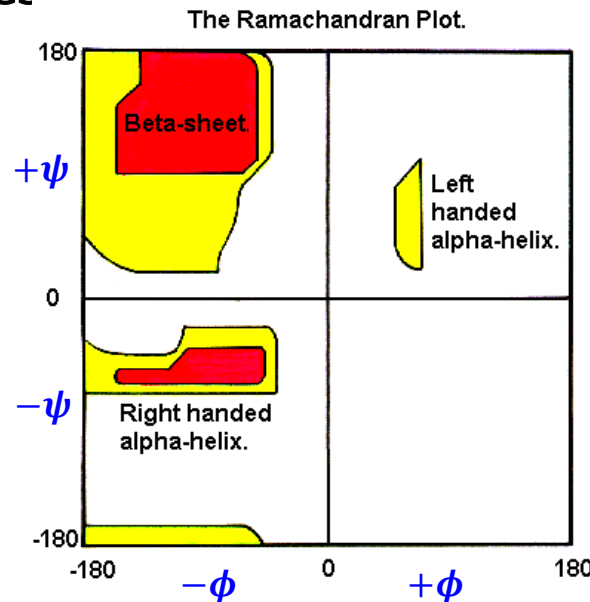
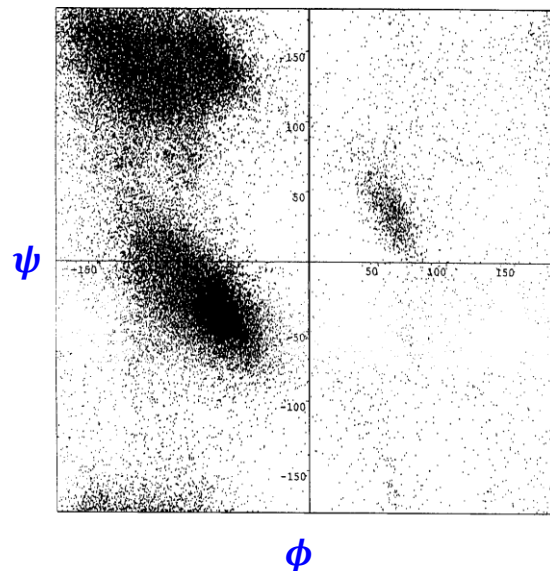


Secondary structure

Ramachandran plot



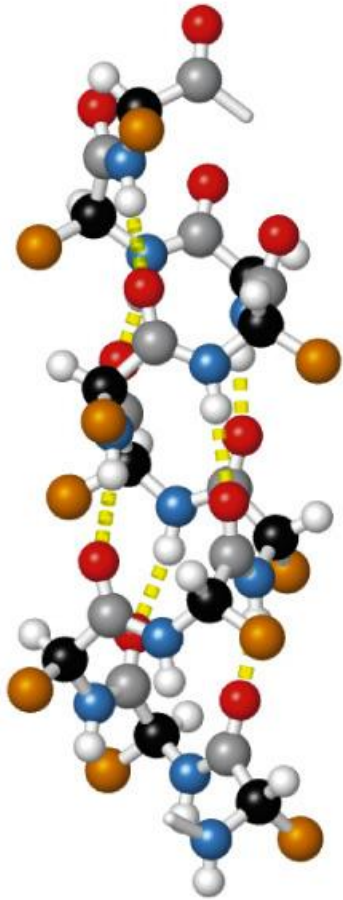
- plot of observed pairs of the ϕ and ψ angles in a collection of known protein structures



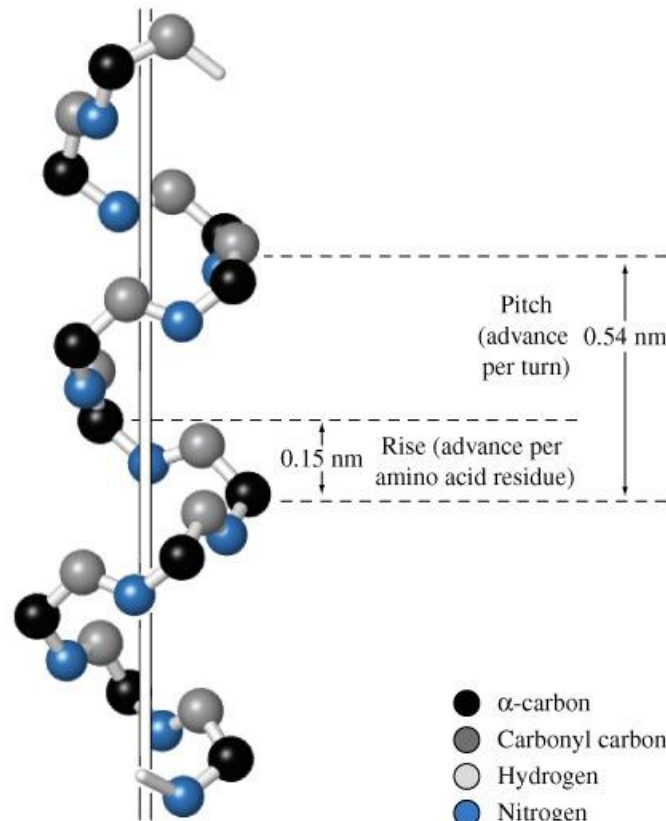
- Describes acceptable ϕ/ψ angles for individual AA's in a polypeptide chain.
- Helps determine what types of secondary structure are present

- The pairs near $\phi = -60^\circ$ and $\psi = -60^\circ$ correspond to helices.
- The pairs near $(-90^\circ, 120^\circ)$ correspond to strands.

α Helix



Right-handed α -helix



Axis

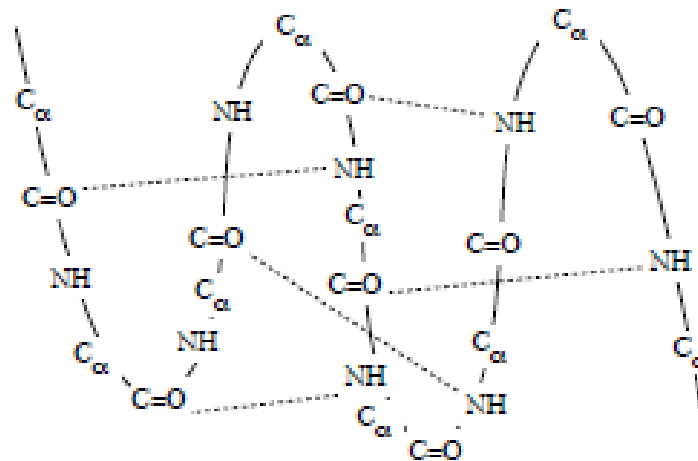
- α -carbon
- Carbonyl carbon
- Hydrogen
- Nitrogen
- Oxygen
- Side chain

- Residues per turn: 3.6
- Rise per residue: 1.5 Ångströms
- $1\text{Å} = 10^{-10}m$
- Rise per turn (pitch): $3.6 \times 1.5\text{Å} = 5.4\text{Å}$
- amino hydrogen H-bonds with carbonyl oxygen located 4 AA's away forms 13 atom loop

Helices

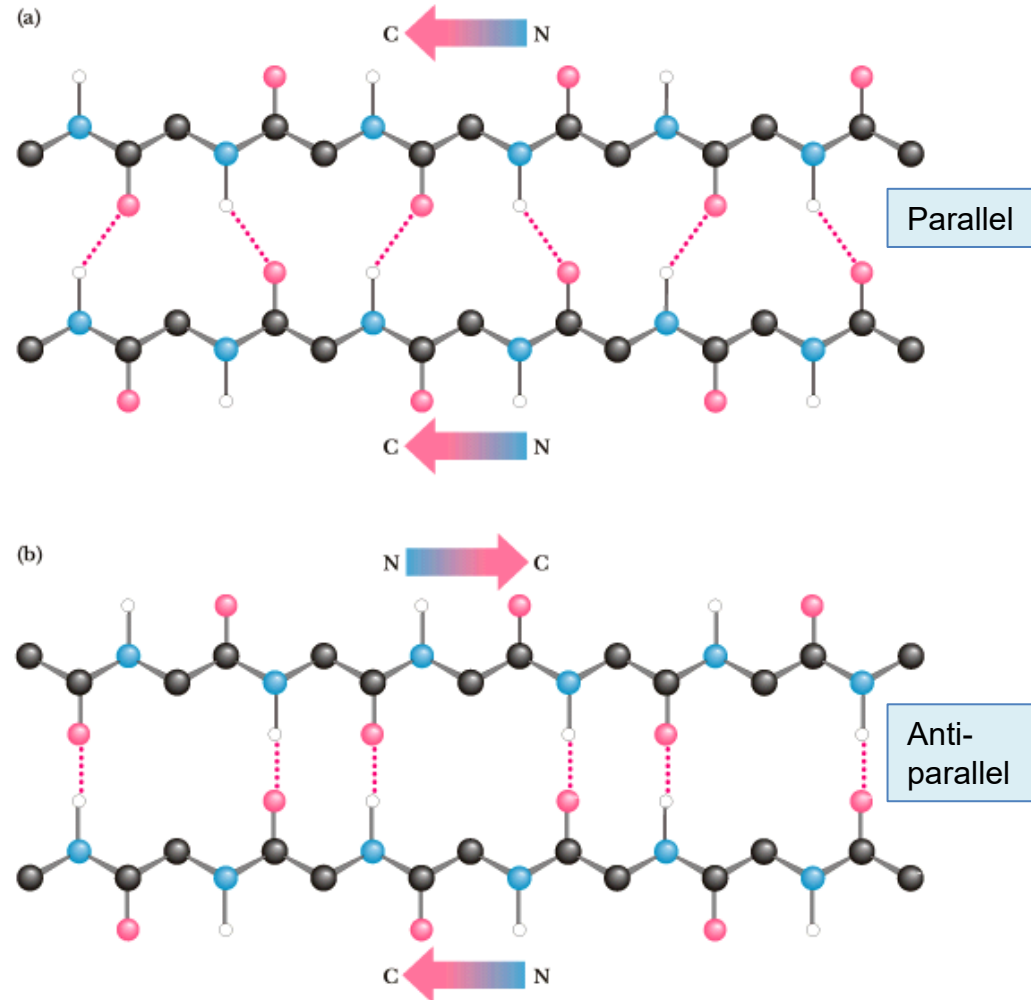


- Helices arise when hydrogen bonds occur between (the C=O group of) the amino acid at position i and (the NH group of) the amino acid at position $i + k$ (with $k = 3, 4$ or 5), for a run of consecutive values of i .
- Most often, $k = 4$ or 5 and the resulting structure is called an α helix, whereas $k = 3$ gives rise to a 3_{10} helix
- α helix:



β Sheets

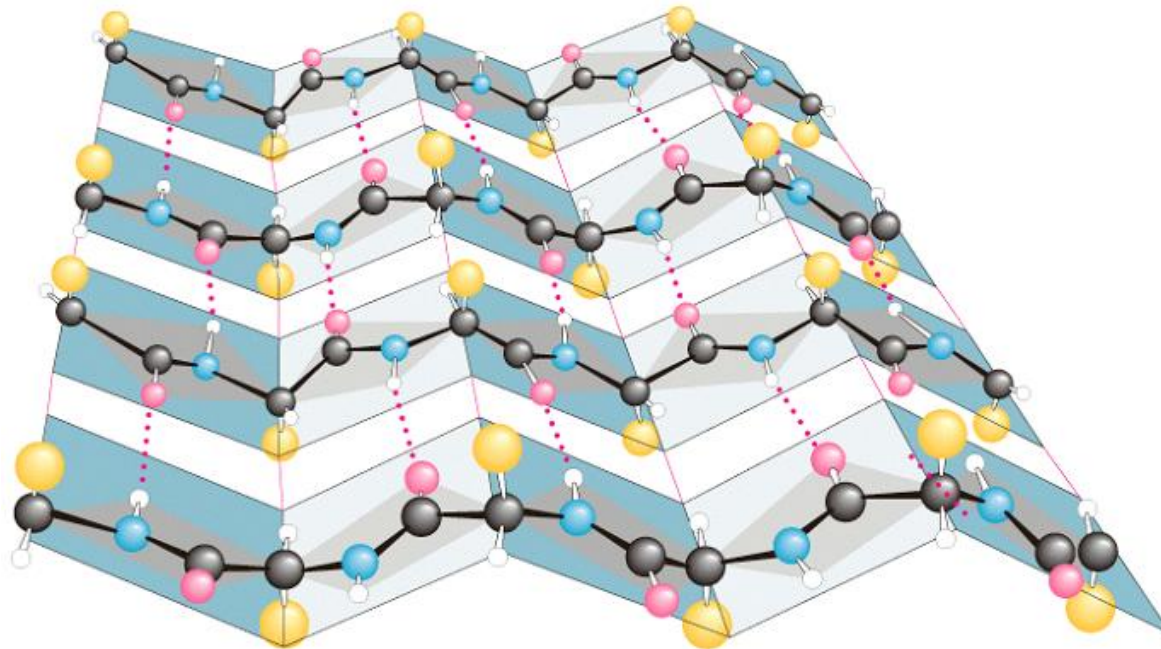
- β -sheets formed from multiple side-by-side beta-strands.
- Can be in parallel or anti-parallel configuration
- Anti-parallel beta-sheets more stable



β Sheets



- Side chains point alternately above and below the plane of the β sheet
- 2- to 15 β -strands/ β -sheet
- Each strand made of ~ 6 amino acids



Loops and turns

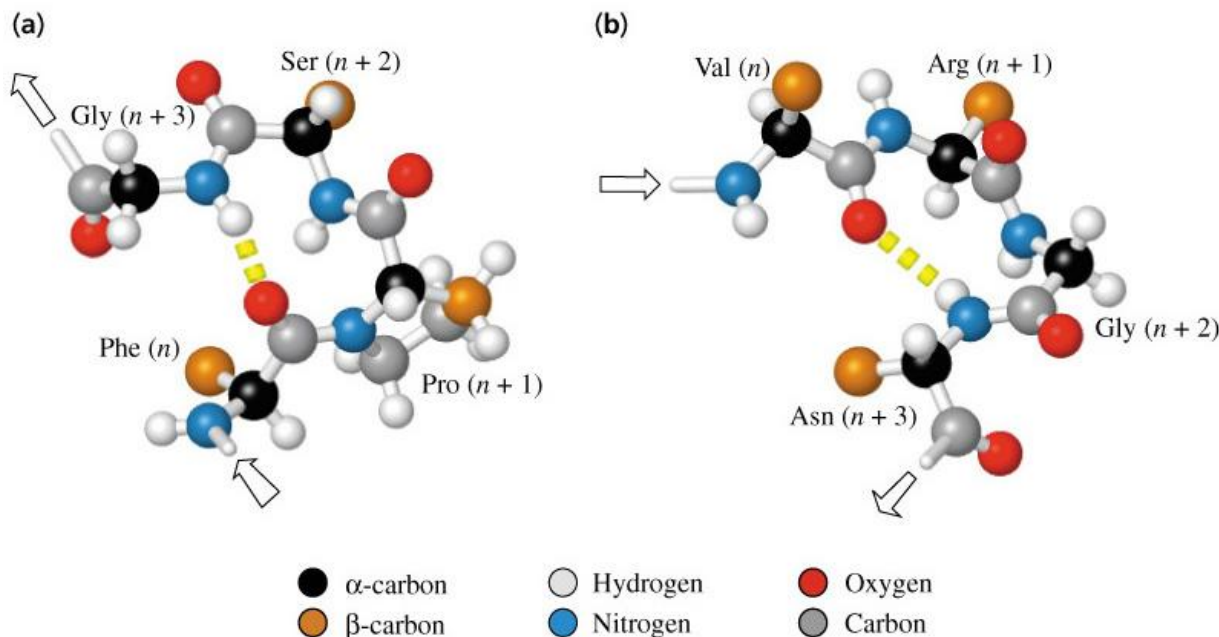


- Loops
 - Loops usually contain hydrophilic residues.
 - Found on surfaces of proteins
 - Connect α helices and β sheets
- Turns
 - Loops with < 5 AA's are called turns
 - β turns are common

β Turns



- allows the peptide chain to reverse direction
- carbonyl C of one residue is H-bonded to the amide proton of a residue three residues away
- proline and glycine are prevalent in beta turns



*A region of secondary structure that is not a helix, a sheet, or recognizable turn is also called a **coil**.*

Outline



1. Proteins
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3. Protein Structure Determination



- X-ray crystallography
 - X-ray: any size, accurate (1–3 Ångström (10^{-10} m)), sometimes hard to grow crystal

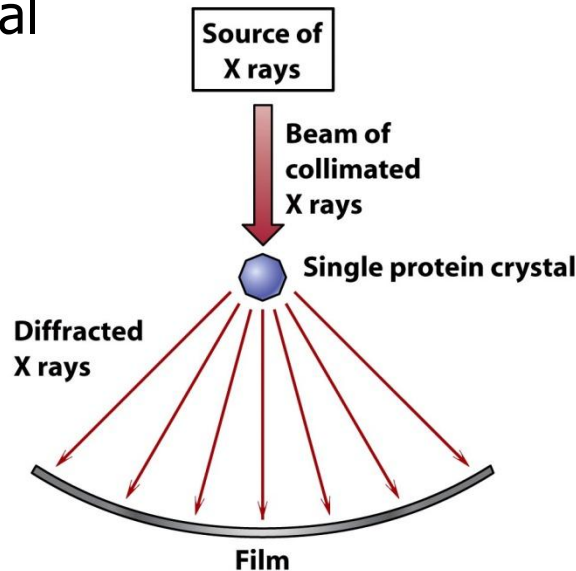


Figure 4-2a Principles of Biochemistry, 4/e
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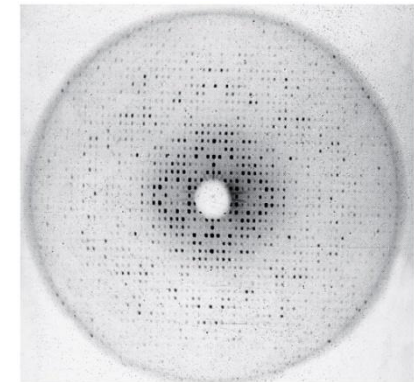
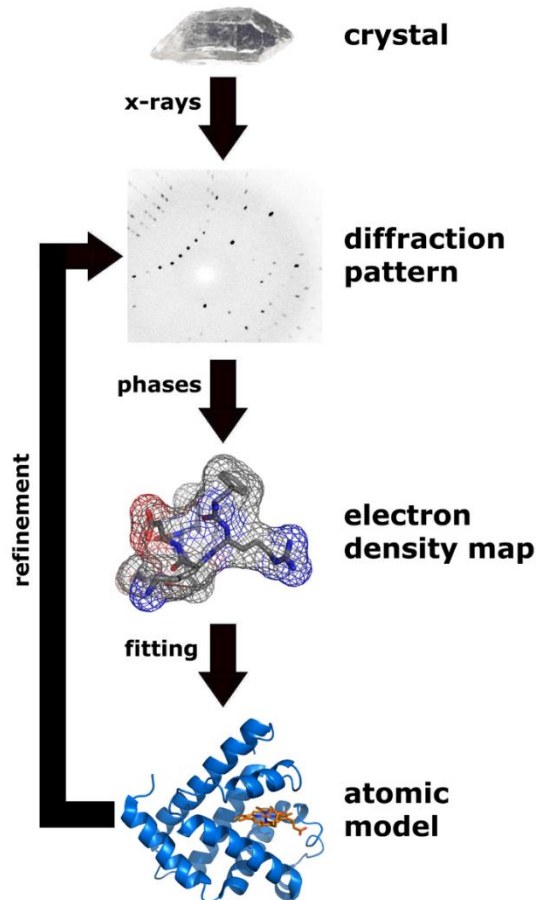


Figure 4-2b Principles of Biochemistry, 4/e
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- Nuclear Magnetic Resonance (NMR) Spectroscopy
 - small to medium size, moderate accuracy, structure in solution

X-ray crystallography

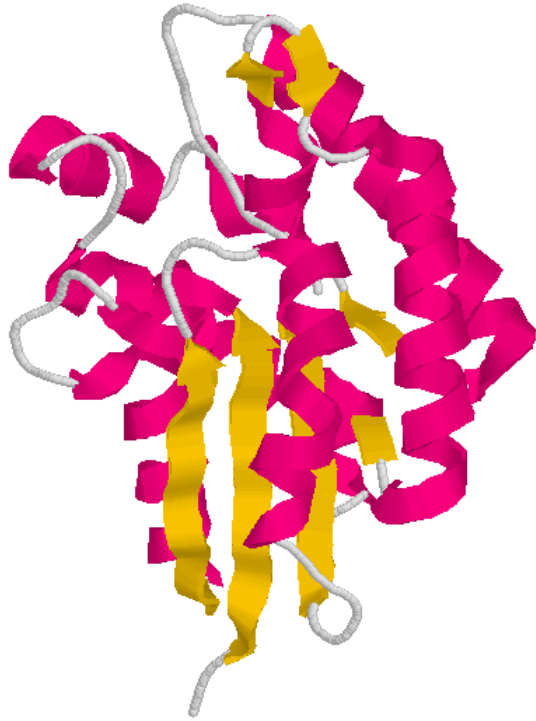


Wikipedia, the free encyclopedia

By Thomas Splettstoesser

<https://commons.wikimedia.org/w/index.php?curid=1248574>

1D: Secondary Structure Prediction



MWLKKFGINLLIGQSV...

Neural Networks
+ Alignments

H – α helix
S – β sheet
L – loop (or coil)

LLLLHHHHLLLSSSSS...

Accuracy: 78%

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How to Use Neural Network to Predict Secondary Structure



1. Create a data set with input sequences (X) and output labels (secondary structures)
2. Encode the input and output to neural network
3. Train neural network on the dataset (training dataset)
4. Test on the unseen data (test dataset) to estimate the generalization performance.

Create a Data Set



- Download proteins from Protein Data Bank
- Select high-resolution protein structures (< 2.5 Ångström, determined by X-ray crystallography)
- Remove proteins with chain-break ($C_{\alpha}-C_{\alpha}$ distance > 4 Ångström)
- Remove redundancy (filter out very similar sequences using BLAST)
- Use [DSSP program](#) (Kabsch and Sander, 1983) to assign secondary structure to each residue.
 - DSSP is a database of secondary structure assignments (and much more) for all protein entries in the Protein Data Bank (PDB). DSSP is also the program that calculates DSSP entries from PDB entries. DSSP does **not** predict secondary structure – it computes secondary structure from 3D structure.

[A series of PDB related databases for everyday needs.

Wouter G Touw, Coos Baakman, Jon Black, Tim AH te Beek, E Krieger, Robbie P Joosten, Gert Vriend.

Nucleic Acids Research 2015 January; 43 (Database issue): D364-D368.]

Train and Test



- Use one data set as training dataset to build neural network model
- Use another data set as test dataset to evaluate the generalization performance of the model
- Sequence similarity of any two sequences in test and training dataset should be less than 25%.

Create Inputs and Outputs for Feed-Forward NN for a Single Sequence



Protein Sequence:

MWLKKF'GINLLIGQSVQTRSWYYCKRA



How to encode the input for each position?
How to encode the output for each position?

SS Sequence:

LLLLHHHHHHHEEEEEHHHHEEEEEELL

H – α helix
E – β sheet – extended strand
L – loop (or coil)

Create Inputs and Outputs for Feed-Forward NN for a Single Sequence



Protein Sequence:

MWLKKF'GINLLIGQSVQTRSWYYCKRA



One-hot-encoding

Use 20 inputs of 0s and 1s for each amino acid
Use 3 inputs to encode the SS alphabet

SS Sequence:

LLLLHHHHHHEEEEEHHHHEEEEEELL

100: Helix

010: Extended strand

001: Loop (or coil)

Similarly for 20 different amino acids

Use a Window to Account for Context



Protein Sequence:

MWLKKFGINLLIGQSVQTRSWYYCKRA



SS Sequence:

LLLLHHHHHHHEEEEEHHHHEEEEEELL

Total number of inputs is **window size $l \cdot 20$** .
 l is a parameter to tune.

Use an Extra Input to Account for N- and C- Terminal Boundary



Protein Sequence:

◇◇◇MWLKKFGINLLIGQSVQTRSWYYCKRA

SS Sequence:

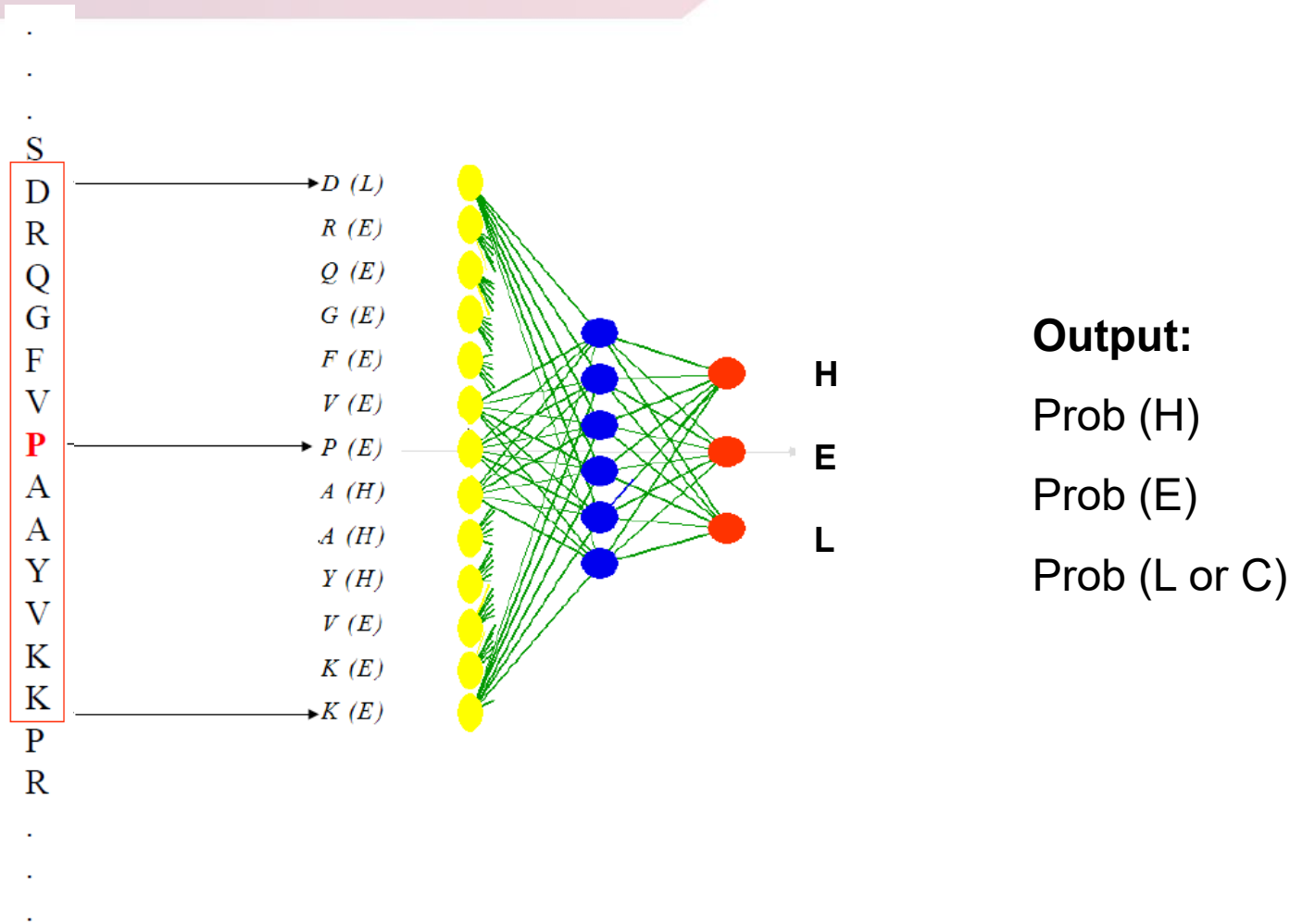


LLLLHHHHHHHEEEEEHHHHEEEEEELL

Add an extra input for each position to indicate if it is out of the boundary of the sequence ('spacer').

Total number of inputs is window size $l \cdot 21$, l is a parameter to tune.

Secondary Structure Prediction (Generation III – Neural Network)



Evolutionary Information is Important



- Single sequence yields accuracy below 70%.
- Use all the sequences in the family of a query sequence can improve accuracy to 78%.
- Structure is more conserved than sequence during evolution. The conservation and variation provides key information for secondary structure prediction.

Second Breakthrough: Evolutionary Information - Profile



	1				50
fyn_human	VTLFVALYDY	EARTEDDLSF	HKGEKFQILN	SSEGDWWEAR	SLTTGETGYI
yrk_chick	VTLFIALYDY	EARTEDDLSF	QKGEKFHIIN	NTEGDWWEAR	SLSSGATGYI
fgr_human	VTLFIALYDY	EARTEDDLTF	TKGEKFHILN	NTEGDWWEAR	SLSSGKTGCI
yes_chick	VTVFVALYDY	EARTTDDLFS	KKGERFQIIN	NTEGDWWEAR	SIATGKTGYI
src_avis2	VTTFVALYDY	ESRTE TDLFS	KKGERLQIVN	NTEGDWWLAH	SLTTGQTGYI
src_avis	VTTFVALYDY	ESRTE TDLFS	KKGERLQIVN	NTEGDWWLAH	SLTTGQTGYI
src_avisr	VTTFVALYDY	ESRTE TDLFS	KKGERLQIVN	NTEGDWWLAH	SLTTGQTGYI
src_chick	VTTFVALYDY	ESRTE TDLFS	KKGERLQIVN	NTEGDWWLAH	SLTTGQTGYI
stk_hydat	VTIFVALYDY	EARISEDLSF	KKGERLQIIN	TADGDWWYAR	SLITNSEGYI
src_rsvpa		ESRIETDLFS	KKRERLQIVN	NTEGTWWLAH	SLTTGQTGYI
hck_human	..IVVALYDY	EAIHHEDLSF	QKGDQMVVLE	ES.GEWWKAR	SLATRKEGYI
blk_mouse	..FVVALFDY	AAVNDRLQV	LKGEKLQVLR	.STGDWWLAR	SLVTGREGYV
hck_mouse	.TIVVALYDY	EAIHREDLSF	QKGDQMVVLE	.EAGEWWKAR	SLATKKEGYI
lyn_human	..IVVALYPY	DGIHPDDLFS	KKGEKMKVLE	.EHGEWWKAK	SLLTKKEGFI
lck_human	..LVIALHSY	EPSHDGDLGF	EKGEQLRILE	QS.GEWKKAQ	SLTTGQEGFI
ss81_yeast	ALYPY	DADDDdeISF	EQNEILQVSD	.IEGRWWKAR	R.ANGETGII
abl_mouse	..LFVALYDF	VASGDNTLSI	TKGEKLRLVG	YnnGEWCEAQ	..TKNGQGWW
abl1_human	..LFVALYDF	VASGDNTLSI	TKGEKLRLVG	YnnGEWCEAQ	..TKNGQGWW
src1_drome	..VVVSLYDY	KSRDESDLSF	MKGDRMEVID	DTESDWWRVV	NLTTRQEGLI
mysd_dicdi	ALYDF	DAESSMELSF	KEGDILTVLD	QSSGDWWDAE	L..KGRRGKV
yfj4_yeast	VALYSF	AGEESGDLPF	RKGDVITILK	ksQNDWWTGR	V..NGREGIF
abl2_human	..LFVALYDF	VASGDNTLSI	TKGEKLRLVG	YNQNGEWSEV	RSKNG.QGWW
tec_human	.EIVVAMYDF	QAAEGHDLRL	ERGQEYLILE	KNDVHWWRAR	D.KYGNEGYI
abl1_caeel	..LFVALYDF	HGVGEEQLSL	RKGDQVRILG	YNKNNEWCEA	RlrLGEIGWV
txk_human	ALYDF	LPREPCNLAL	RRAEEYLILE	KYNPHWWKAR	D.RLGNEGLI
yha2_yeast	VRRVRALYDL	TTNEPDELSF	RKGDVITVLE	QVYRDWWKGA	L..RGNMGI
abp1_sacex	AEYDY	EAGEDNELTF	AENDKIINIE	FVDDDWLGE	LETTGQKGLF

B. Rost, 2005

How to Find Homologous Sequences and Generate Alignments

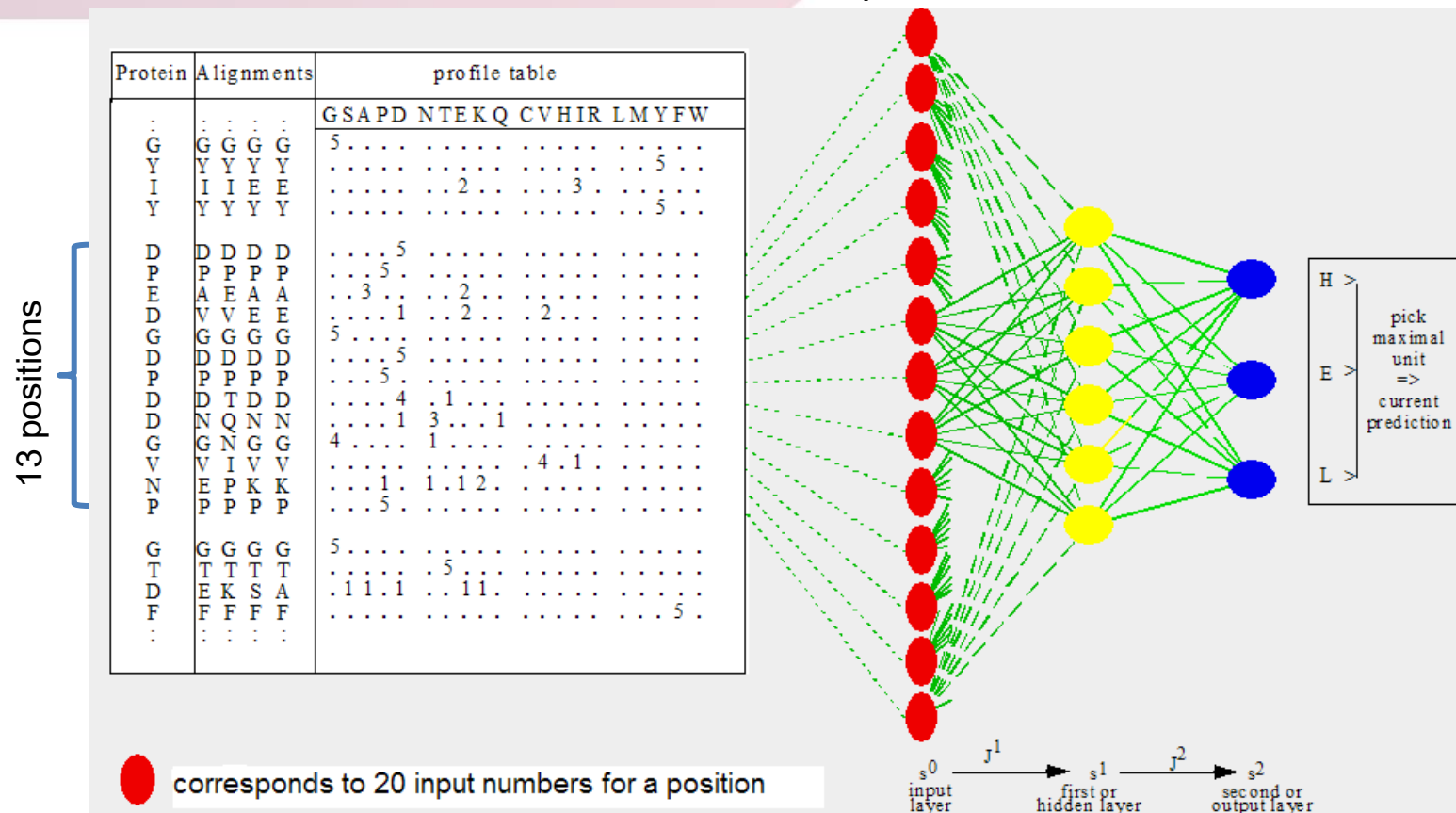


Position Specific Iterated BLAST

- Use PSI-BLAST to search a query sequence against the very large non-redundant protein sequence database (NR database, compiled at NCBI)
- Combine the pairwise alignment between the query sequence and other sequences into a multiple sequence alignment using the query sequence as the center.

PHD Approach

Dr. Pierre Baldi and his team at the University of California, Irvine



Comments: frequency is normalized into probability and sequence needs to be weighted.

Reference: Rost and Sander. Proteins, 1994.

Second Neural Network to Smooth Output Predictions



- Raw output from one neural network may contain weird predictions such as helix of length 1. But minimum length is 2.
- So, use another neural network to smooth output. The inputs are a window of predicted secondary structure. The outputs are the true secondary structures.
- The second neural network makes the predictions more protein-like.

PHD Approach



Local alignment

13 adjacent positions

Global statistics
on whole protein



Input local in the sequence – for each residue (13) use

- 20 values from the profile
- 1 'spacer' – yes/no position out of the sequence
- 2 number of insertions and deletions in the alignment
- 1 'cons' conservation weight

Global statistics

- 20 amino acid composition
- 4 protein length ($\leq 60, \leq 120, \leq 240, > 240$)
- 8 distances of the window from the protein end ($\leq 40, \leq 30, \leq 20, \leq 10$)



First level NN:

- Hidden layer and
- 3 outputs (helix, strand, other) for the central residue



Second level NN:

- Input for each residue (13)
 - 3 output of the first level
 - 'spacer'
 - 'cons'
- Hidden layer and output similar to the first level

PhD Approach



- The second level introduces a correlation between adjacent residues, otherwise, e.g., too short helices are outputted
- The distribution of examples is uneven
 - 32% of the residues in helices,
 - 21% in strand, and
 - 47% in loop
- A balanced training is used – it improved results for less frequent states but has not decreased accuracy for high frequency residues
⇒ lower overall accuracy
- Final decision – a **jury**: an arithmetic average over 4 differently trained networks: all combinations of the first level NN with balanced/unbalanced training and second level NN with balanced/unbalanced training
- Final prediction = unit with the maximal value

PSI-PRED Approach



- PSI-PRED does not use probability matrix instead it uses another kind of profile: Position Specific Scoring Matrix (PSSM) generated by PSI-BLAST during sequence search.
- The weighting of the sequences is done implicitly by PSI-BLAST.
- The raw PSSM is transformed into values within $[0,1]$ using sigmoid function.

What is the difference between probability matrix and PSSM?

Reference: Jones, Journal of Molecular Biology, 1999.

PSI-PRED Approach



- PSI-PRED does not use probability matrix instead it uses another kind of profile: Position Specific Scoring Matrix (PSSM) generated by PSI-BLAST during sequence search.
- The weighting of the sequences is done implicitly by PSI-BLAST.
- The raw PSSM is transformed into values within $[0,1]$ using sigmoid function.

What is the difference between probability matrix and PSSM?

A probability matrix represents the probabilities of observing different residues at each position in a multiple sequence alignment, while a PSSM represents the scores that indicate the evolutionary conservation and amino acid preferences at each position in the alignment

Reference: Jones, Journal of Molecular Biology, 1999.

PSI-PRED Input



Position-based scoring matrix used

A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
-3	-4	-4	-4	-3	-4	-4	-4	-2	-1	-1	-4	-1	8	-5	-3	-3	0	2	-2
0	-1	-1	3	-4	3	4	1	-1	-4	-4	0	-3	-4	-2	-1	-2	-4	-3	-3
0	-1	2	1	-3	4	0	-1	-2	-4	-3	1	-2	-4	-2	2	0	-4	-3	-3
-2	-3	-4	-5	-2	-3	-4	-6	-4	0	6	0	0	-1	-4	-3	-2	-4	-2	0
0	-3	-1	-2	-3	0	-2	4	-3	-3	0	-2	-2	-4	-3	3	1	-4	-4	-3
0	2	0	4	-4	1	2	1	-2	-4	-4	0	-3	-4	-3	1	-2	-5	-4	-4
-1	5	3	-2	-4	-1	-1	1	-2	-1	-4	1	-3	-4	-3	1	-2	-5	-4	-4
-2	-3	-4	-5	-3	-3	-4	-5	-4	3	4	-1	1	2	-4	-3	-2	-3	-1	0
-2	3	2	-2	-4	2	1	-3	-2	-3	-3	1	1	-4	-3	2	1	-4	-3	-1
0	2	3	1	-4	0	0	0	-2	-4	-4	1	-3	-4	-3	2	0	-5	-4	-4
5	-3	-3	-3	-2	-3	-3	-2	-3	1	-2	-3	-2	1	-3	0	1	-4	-2	0
-1	-4	-5	-5	-3	-4	-4	-5	-4	3	3	-4	2	3	-5	-3	-2	5	-1	2
0	3	3	0	-4	3	0	1	-2	-4	-4	1	-3	-4	-3	1	-1	-4	-3	-4
-1	0	1	0	-4	1	-1	-1	-2	-4	-3	5	-2	0	-3	0	-2	-4	0	-3
-2	-3	-1	-5	-3	-3	-4	-5	-4	3	4	0	4	2	-4	-3	-2	-3	-2	0
0	3	0	-2	-3	-1	0	0	-2	0	0	1	0	-1	-3	2	0	-4	-3	0
-1	1	3	-2	-4	0	-2	4	-2	-4	-4	0	-3	0	-3	0	0	-3	0	-4

Window of
15 rows

Sigmoid
transformation

$$\frac{1}{1 + e^{-x}}$$

A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
0.4	0.3	0.3	0.3	0.2	0.9	0.3	0.3	0.4	0.4	0.4	0.3	0.4	0.9	0.1	0.4	0.4	0.5	0.7	0.4
0.3	0.2	0.3	0.8	0.4	0.3	0.7	0.1	0.6	0.2	0.4	0.3	0.5	0.2	0.1	0.4	0.8	0.2	0.3	0.2
0.1	0.1	0.4	0.3	0.5	0.1	0.1	0.3	0.1	0.1	0.4	0.2	0.4	0.9	0.3	0.4	0.4	0.9	0.3	0.6
0.6	0.3	0.3	0.1	0.3	0.5	0.5	0.2	0.1	0.4	0.4	0.3	0.6	0.9	0.1	0.5	0.1	0.5	0.7	0.4

Reference: Jones, Journal of Molecular Biology, 1999.

PSI-PRED



Raw profile from PSI-BLAST Log File

Position-based scoring matrix used

A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
-3	-4	-4	-4	-3	-4	-4	-4	-2	-1	-1	-4	-1	8	-5	-3	-3	0	2	-2
0	-1	-1	3	-4	3	4	1	-1	-4	-4	0	-3	-4	-2	-1	-2	-4	-3	-3
0	-1	2	1	-3	4	0	-1	-2	-4	-3	1	-2	-4	-2	2	0	-4	-3	-3
-2	-3	-4	-5	-2	-3	-4	-6	-4	0	6	0	0	-1	-4	-3	-2	-4	-2	0
0	-3	-1	-2	-3	0	-2	4	-3	-3	0	-2	-2	-4	-3	3	1	-4	-4	-3
0	2	0	4	-4	1	2	1	-2	-4	-4	0	-3	-4	-3	1	-2	-5	-4	-4
-1	5	3	-2	-4	-1	-1	1	-2	-1	-4	1	-3	-4	-3	1	-2	-5	-4	-4
-2	-3	-4	-5	-3	-3	-4	-5	-4	3	4	-1	1	2	-4	-3	-2	-3	-1	0
-2	3	2	-2	-4	2	1	-3	-2	-3	-3	1	1	-4	-3	2	1	-4	-3	-1
0	2	3	1	-4	0	0	0	-2	-4	-4	1	-3	-4	-3	2	0	-5	-4	-4
5	-3	-3	-3	-2	-3	-3	-2	-3	1	-2	-3	-2	1	-3	0	1	-4	-2	0
-1	-4	-5	-5	-3	-4	-4	-5	-4	3	3	-4	2	3	-5	-3	-2	5	-1	2
0	3	3	0	-4	3	0	1	-2	-4	-4	1	-3	-4	-3	1	-1	-4	-3	-4
-1	0	1	0	-4	1	-1	-1	-2	-4	-3	5	-2	0	-3	0	-2	-4	0	-3
-2	-3	-1	-5	-3	-3	-4	-5	-4	3	4	0	4	2	-4	-3	-2	-3	-2	0
0	3	0	-2	-3	-1	0	0	-2	0	0	1	0	-1	-3	2	0	-4	-3	0
-1	1	3	-2	-4	0	-2	4	-2	-4	-4	0	-3	0	-3	0	0	-3	0	-4

Window of 15 rows

Sigmoid transformation

$$\frac{1}{1 + e^{-x}}$$

A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
0.4	0.3	0.3	0.3	0.2	0.9	0.3	0.3	0.4	0.4	0.4	0.3	0.4	0.9	0.1	0.4	0.4	0.5	0.7	0.4
0.3	0.2	0.3	0.8	0.4	0.3	0.7	0.1	0.6	0.2	0.4	0.3	0.5	0.2	0.1	0.4	0.8	0.2	0.3	0.2
0.1	0.1	0.4	0.3	0.5	0.1	0.1	0.3	0.1	0.1	0.4	0.2	0.4	0.9	0.3	0.4	0.4	0.9	0.3	0.6
0.6	0.3	0.3	0.1	0.3	0.5	0.5	0.2	0.1	0.4	0.4	0.3	0.6	0.9	0.1	0.5	0.1	0.5	0.7	0.4
.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.

+1 C or N terminus

15 x 20 scaled inputs to 1st network

1st Network
315 inputs
75 hidden units
3 outputs

Window of 15 x 3
outputs fed to 2nd
network

2nd Network
60 inputs
60 hidden units
3 outputs

Final 3-state
Prediction

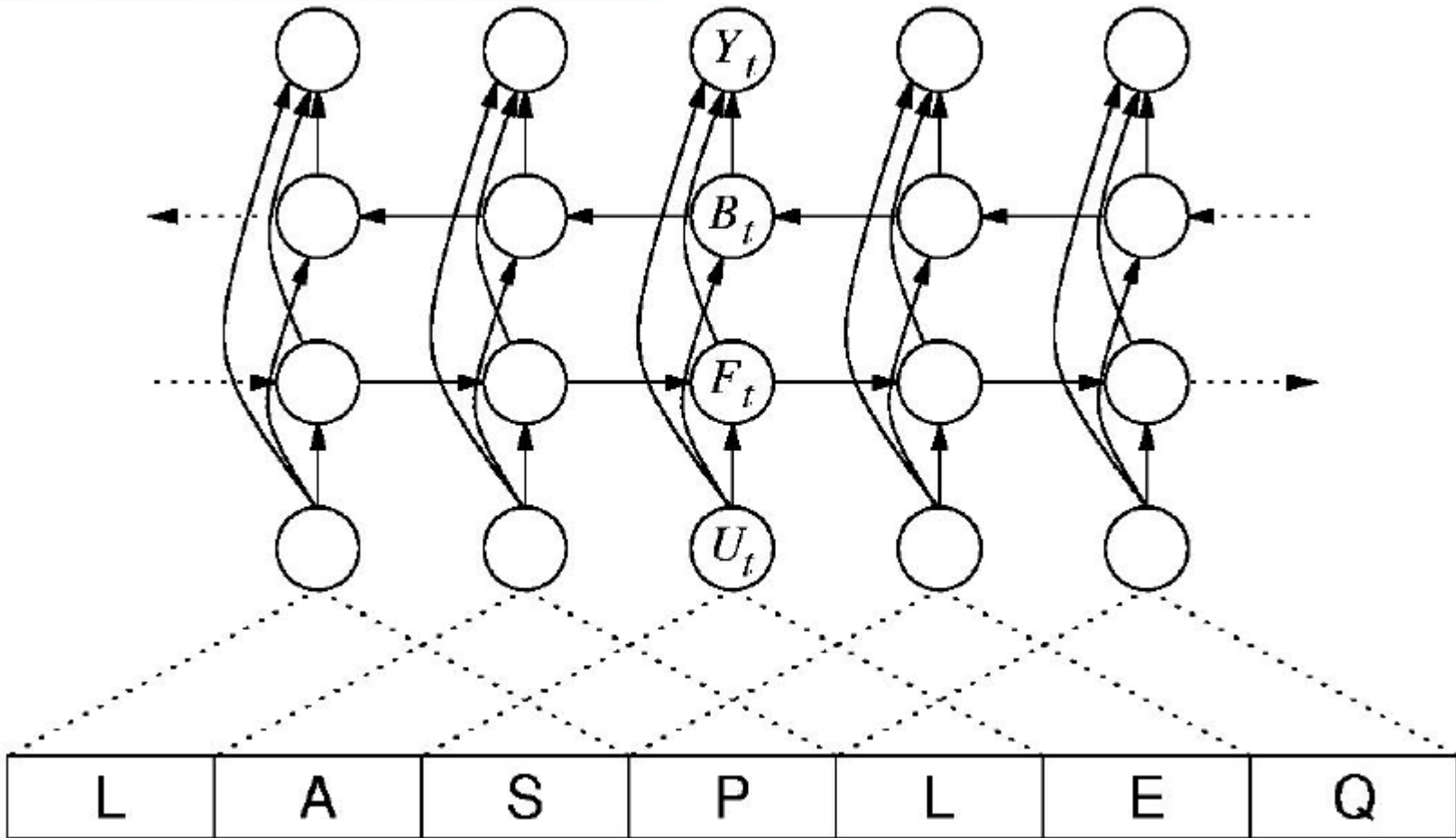
SSpro Approach



- SSpro uses probability matrix as inputs
- SSpro uses an information theory approach to weight sequences
- The main novelty of SSpro is to use 1-Dimensional Recurrent Neural Network (1D-RNN)

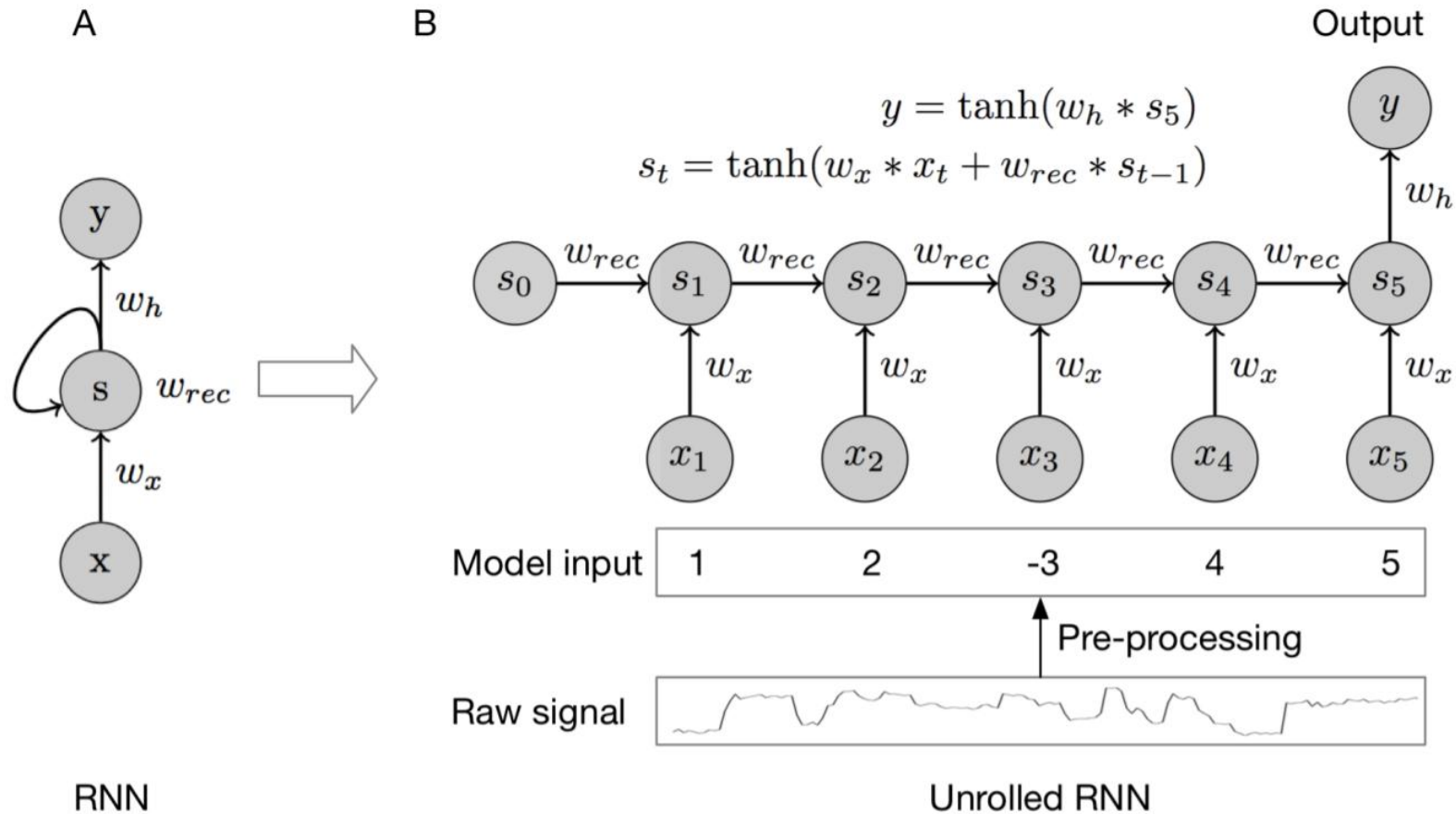
Pollastri et al.. Proteins, 2002.

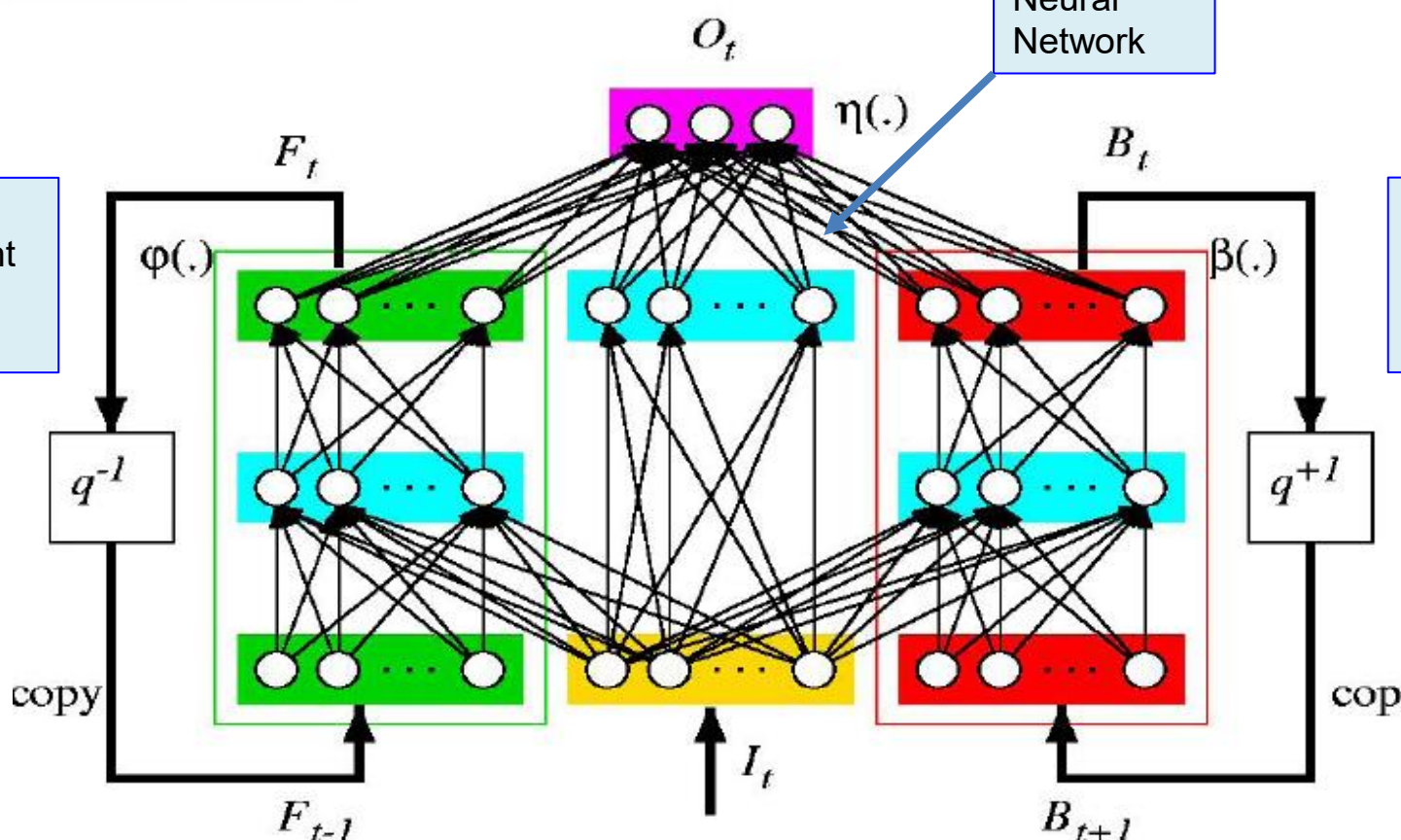
Bi-directional Input Output Hidden Markov Model for SS Prediction



Baldi, 2004

RNN





April 7, 2026

Advantage and Disadvantages of SSpro



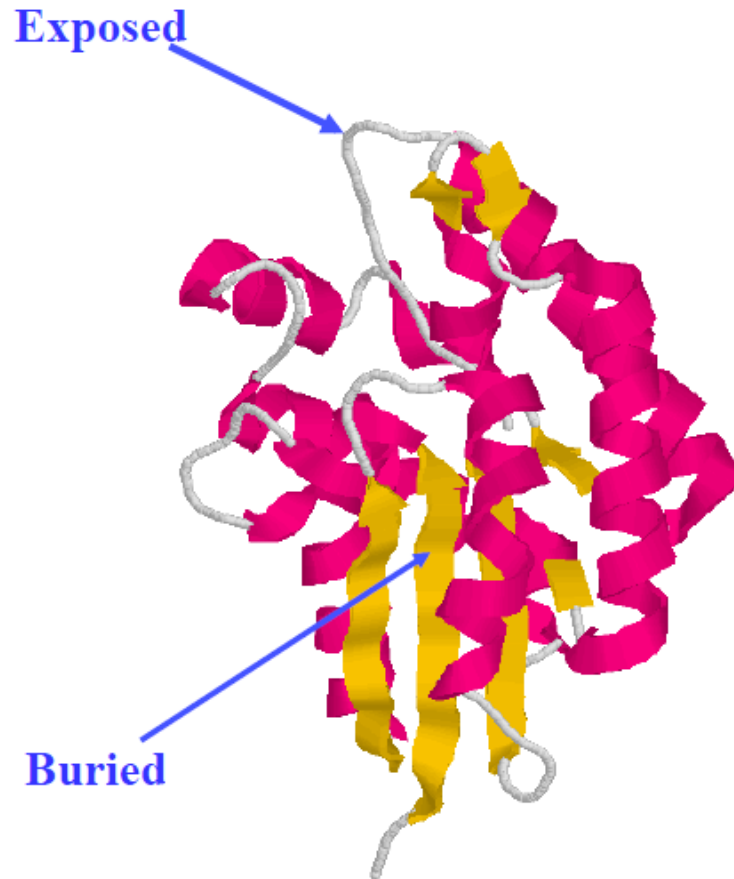
- Directly take a sequence with variable length as inputs.
- Hopefully can utilize more information than a fixed-window approach
- More complex, thus harder to implement than feed-forward neural network.

Outline



1. Proteins
2. Secondary structure
3. Protein structure determination
4. Using neural networks for protein structure prediction
5. **Predicting solvent accessibility, disordered region, contact map**

1D: Solvent Accessibility Prediction



MWLKKFGINLLGAQVSBG...



Neural networks+alignments

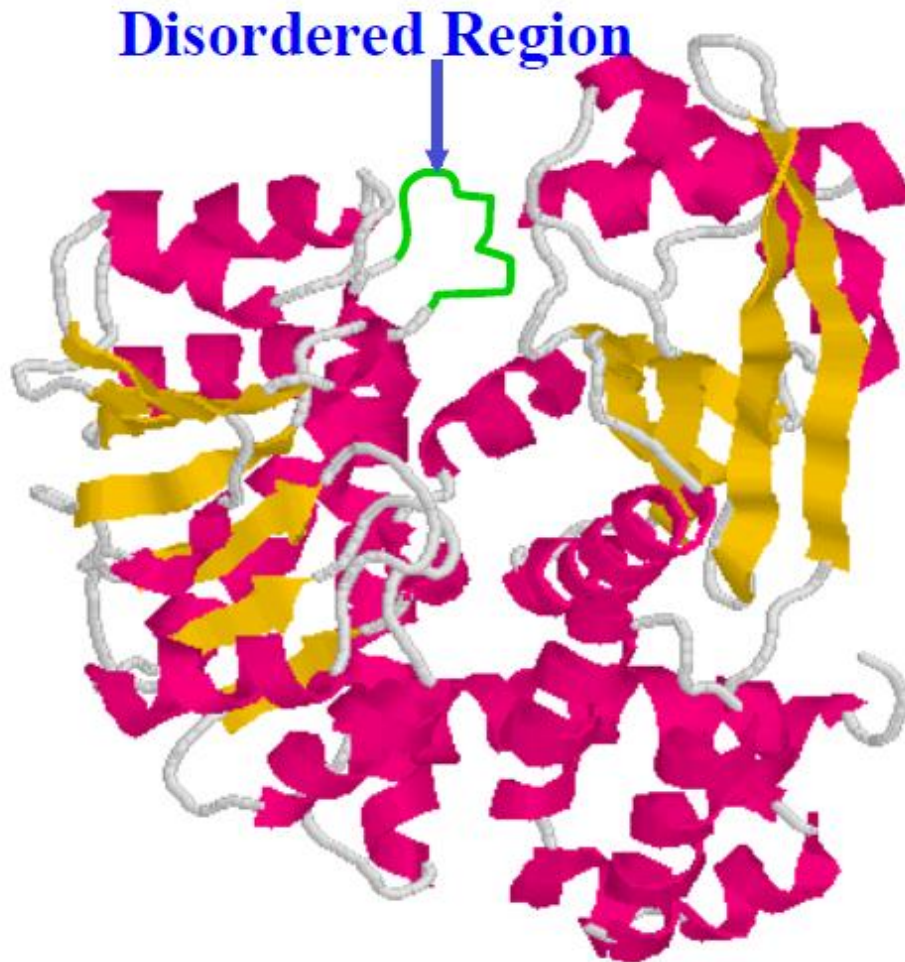


eeeeeebbbbbbbbbeeeebb...

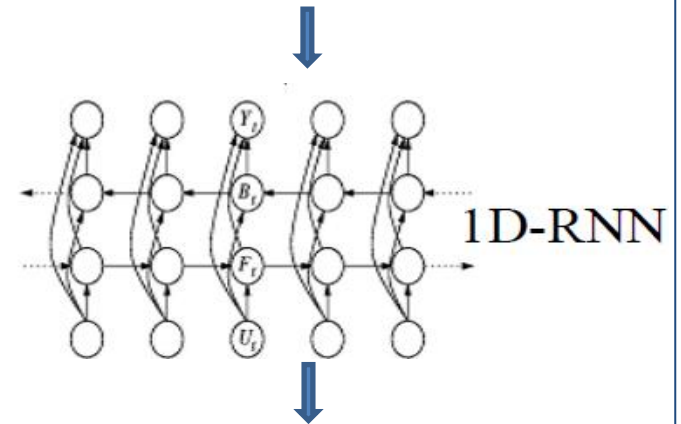
Accuracy: 79%

Pollastri et al. *Proteins*, 2002/
Cheng et al. *Nucleic Acid
Research*, 2005

1D: Disordered Region Prediction Using Neural Networks



MWLKKFGINLLGAQVSBG...



oooooddddddooooooo...

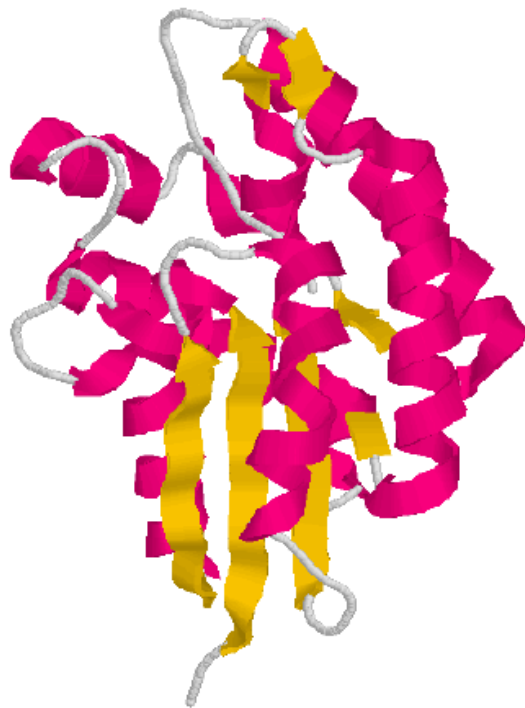
93% TP at 5% FP

Cheng, Sweredoski, Baldi. *Data Mining and Knowledge Discovery*, 2005

2D: Contact Map Prediction



3D Structure

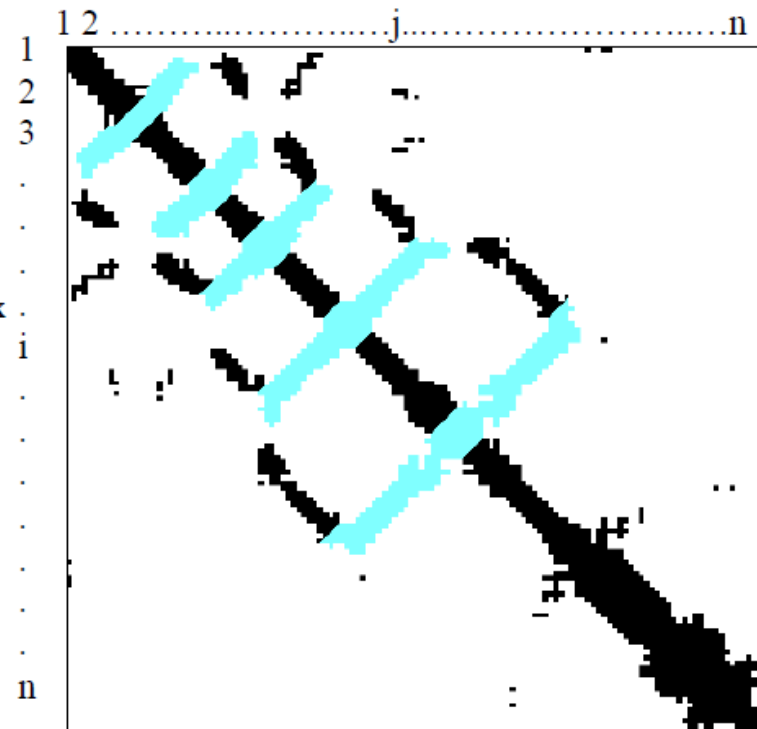


2D-Recursive
Neural Network



Support Vector
Machine

2D Contact Map



Distance Threshold = $8\text{\AA}$

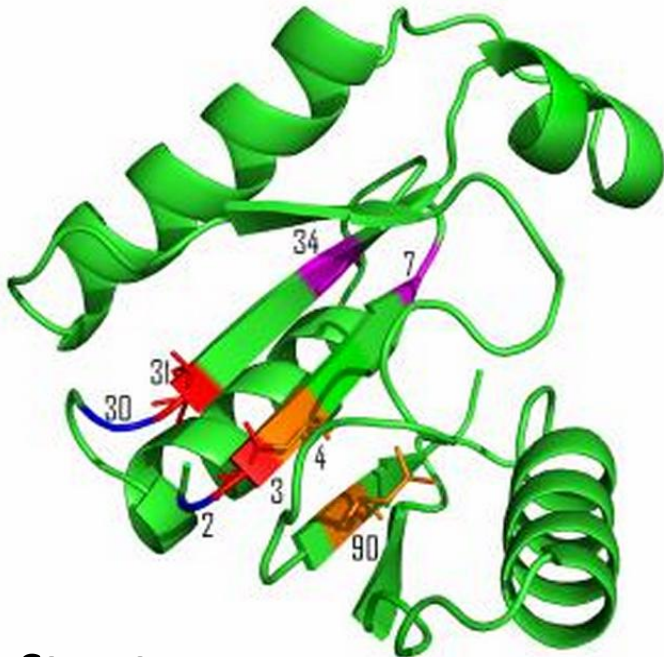
Cheng, Randall, Sweredoski, Baldi. *Nucleic Acid Research*, 2005
Cheng and Baldi. *BMC Bioinformatics*, 2007.

Residue-Residue Contact Prediction



1D Sequence

SDDEVYQYIVSQVKQYGI EPAELLSRKYGDKAKYHLSQRW



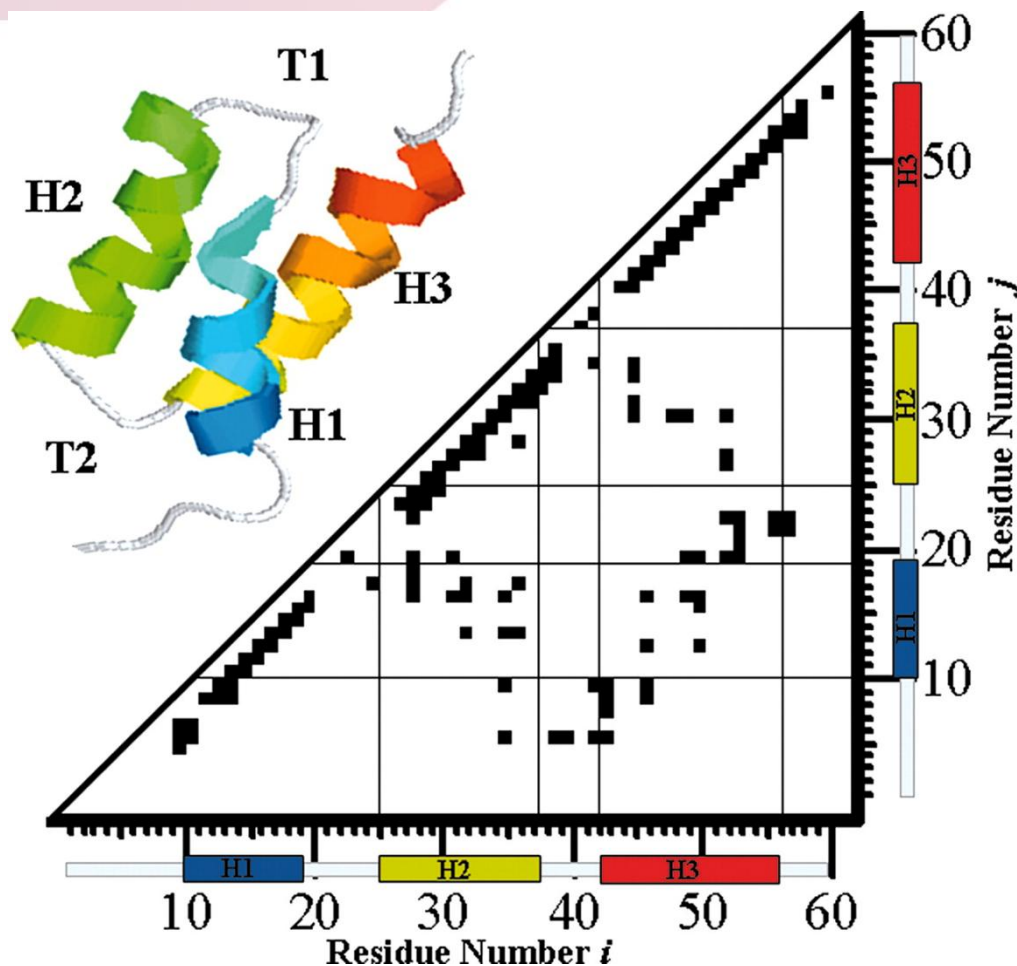
3D Structure

Objective:

Predict if two residues (i, j) are in contact (spatially close), i.e.
 $distance(i, j) < 8 \text{ \AA}$

Eickholt & Cheng,
2012

Visualization of a Contact Map



A Binary Classification Problem



i SDDEVYQYI**V**SQVKQYGI EPCSAELLSRKYG DKAK**Y**HLSQRW j

Residue identity, secondary structure, solvent accessibility, ...

A Vector of ~400 Features
(numbers between 0 and 1)

Probability that **V** and **Y** are in contact?

Cheng & Baldi, 2007; Tegge et al., 2009; Eickholt & Cheng, 2012

YQYIVSQVK

GDKAKYHLSQ

3 numbers

2 numbers

A 3D ribbon diagram of a protein structure. The protein is composed of several magenta helices, yellow strands, and grey coils. Three blue arrows point to specific elements: 'Helix' points to a magenta helix, 'Coil' points to a grey coil, and 'Strand' points to a yellow strand.

Exposed	10
Buried	01

$25 \cdot 18 = 400$ features for a pair (i, j)

AlphaFold



- AlphaFold demonstrated that AI could accurately predict the shape of a protein, at scale and in minutes, down to atomic accuracy.
- novel neural network architectures and training procedures based on the evolutionary, physical and geometric constraints of protein structures
 - to jointly embed multiple sequence alignments (MSAs) and pairwise features, a new output representation and associated loss that enable accurate end-to-end structure prediction,
 - a new equivariant attention architecture, use of intermediate losses to achieve iterative refinement of predictions, masked MSA (multiple sequence alignment) loss to jointly train with the structure, learning from unlabelled protein sequences using self-distillation and self-estimates of accuracy.
- Jumper, J., Evans, R., Pritzel, A. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
<https://doi.org/10.1038/s41586-021-03819-2>

AlphaFold 3 – article in review



- Abramson, J., Adler, J., Dunger, J. *et al.* Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* (2024).
<https://doi.org/10.1038/s41586-024-07487-w>
- It is considered so important that the paper is not finalized, just “accelerated preview”

AlphaFold 3 – article in review



- diffusion-based architecture, which is capable of joint structure prediction of complexes, including proteins, nucleic acids, small molecules, ions, and modified residues
- far greater accuracy on protein-ligand interactions than the state-of-the-art docking tools
- much higher accuracy on protein-nucleic acid interactions than nucleic-acid-specific predictors, and
- significantly higher antibody-antigen prediction accuracy than AlphaFold-Multimer v2.3
- high accuracy modeling across biomolecular space within a single unified deep learning framework
- *“For the interactions of proteins with other molecule types we see at least a 50% improvement compared with existing prediction methods, and for some important categories of interaction we have doubled prediction accuracy.”*

AlphaFold 3 – article in review



Scientists can access the majority of its capabilities, for free, through our newly launched [AlphaFold Server](#), an easy-to-use research tool. To build on AlphaFold 3's potential for drug design, [Isomorphic Labs](#) is already collaborating with pharmaceutical companies to apply it to real-world drug design challenges and, ultimately, develop new life-changing treatments for patients.

AlphaFold 3



- Free non-commercial access [AlphaFold Server](#)
- [Isomorphic Labs](#) is already collaborating with pharmaceutical companies to apply it to real-world drug design challenges and, ultimately, develop new life-changing treatments for patients.